

Topics in Machine Learning

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Abstract

Machine learning, which is predominantly a branch of artificial intelligence, has gained significant attention in the digital era. Machine learning algorithms are used in many applications that we make use of daily. These algorithms are used for various purposes like data mining, image processing, predictive analytics, etc. to name a few. The main advantage of using machine learning is that, once an algorithm learns what to do with data, it can do its work automatically. In this paper, I intend to do a review of various machine learning algorithms which are most frequently used and therefore are the most popular ones.

Introduction

Machine learning algorithms are used almost in everything that we use nowadays to make our life easier in this digital era. Machine Learning algorithms are used in wide variety of fields such as : pattern recognition, virtual personal assistants (like Google), computer games, robotics, traffic prediction, online transportation network (e.g. estimating surge price in peak hour by Pathao app), product recommendation, share market prediction, medical diagnosis, online fraud prediction, E-mail spam filtering, crime prediction through video surveillance system, social media services (face recognition in Facebook).

In this paper I discussed about few frequently used machine learning algorithms and their applications and how to use them in real life scenario. This paper has 4 chapter. A short description of What I discussed in each of the chapters is given below:

Introduction to Machine Learning: In this chapter, we will define machine learning, it's scopes and limitations, we'll briefly talk about supervised and unsupervised learning algorithm and their classifications and an example for each learning type. And then we'll define simple and multiple linear regression and non-linear regression model with an example for each model.

Linear Regression: In this chapter, we will define hypothesis function and cost function of a linear regression model, we'll visualize the cost function in 3D space and to estimate the parameters of hypothesis function we'll use direct method and gradient descent method, we'll discuss how gradient descent algorithm behaves for small and large learning rate and we'll discuss whether the slope is positive or negative how the gradient descent algorithm always brings the parameters β_j towards minimum, then we'll discuss about the accuracy and relative error of the linear regression model and finally we'll give some real life examples which follow linear regression model.

Logistic Regression: We will give an introduction of Logistic Regression and briefly discuss about different types of Logistic Regression and then, we will show what are the differences between Linear and Logistic Regression model using graphs and why Linear Regression is not a good fit for classification problem and then we will define hypothesis function and cost function for Logistic Regression and we will show what will happen if we put Logistic Regression hypothesis function in Linear Regression cost function, then we will discuss about linear and non-linear decision boundary with some examples and in the end we will show some applications of Logistic Regression.

Regularization: We will define regularization and we will discuss about bias and variance which are most frequently used terms in regularization, we will discuss about over-fitting, under-fitting using graphs and reasons for why they occur and how to avoid them, and then we will show how we can get a good-fit model from an under-fit or over-fit model using ℓ_1 and ℓ_2 regularization and how we can get the value for tuning parameter λ using L-curve method and finally we will discuss about the differences between ℓ_1 and ℓ_2 regularization.

Chapter 1

Introduction to Machine Learning

1.1 What is Machine Learning(ML)?

Machine learning (ML) is the study of computer algorithms that can improve automatically through experience and by the use of data[6]. Machine learning algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to do so[13]. In 1959, Arthur Samuel, a pioneer in the field of machine learning (ML) defined it as the “field of study that gives computers the ability to learn without being explicitly programmed”.

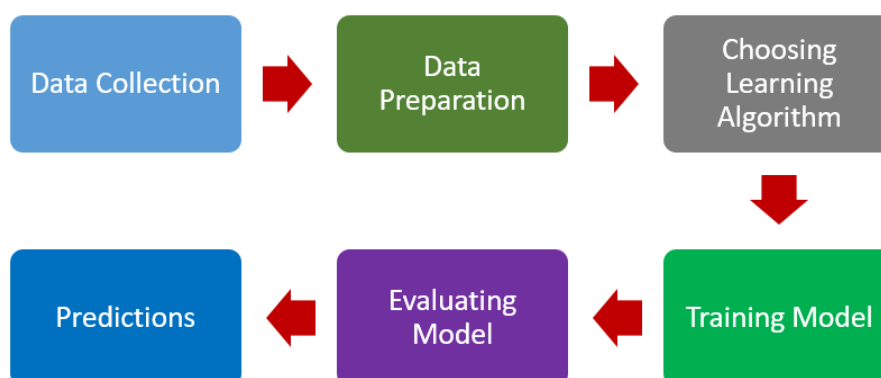


Figure 1.1: Machine Learning workflow

1.2 Scopes of Machine Learning

Machine learning algorithms are used in a wide variety of applications, such as in

- **Speech recognition:** Machine learning is assisting in the evolution of speech recognition, which aims to convert spoken words into text[11]. It is useful and advantageous in a wide range of fields. Many people are unable to communicate due to language problems. Speech recognition software enables users to operate computers by speaking commands into the device. It can translate between languages and recognize human speech.
- **Computer vision:** A computers ability to understand images and videos is referred to as Computer vision. It mimics the capability of human vision by acquiring,

processing, and analyzing real-world data and synthesizing them into useful information. It copies the capability of human vision by gathering, procession, and analyzing real-world data and creating meaningful information from them. Computer vision and machine learning are frequently combined to efficiently collect, process, and interpret collected visual data.

- **Pattern recognition:** The technique of identifying patterns with the aid of a machine learning algorithm is called pattern recognition. The classification of data based on previously acquired knowledge or on statistical data extrapolated from patterns and/or their representation is known as pattern recognition. Pattern recognition's application potential is one of its key features[10].

Examples: Speaker identification, Multimedia Document Recognition, Speech Recognition.

- **Finance:** In the financial sector, machine learning algorithms are used to detect fraudulent activity, automate the trading process and provide investors with financial advisory services[4].

1.3 Limitations of Machine Learning

While developing a machine learning model we might face different kinds of challenges such as

- **Irrelevant features**-which could turn a good machine learning model to a bad model because if the data-set contains irrelevant features then the model won't be able to predict accurately what it was supposed to predict.
- **Insufficient training data**-if machine is not provided with enough data then it won't be able to build a model properly and will be unable to detect the true trend between independent and dependent variables.

1.4 Types of ML model

Machine learning models are grouped into different types, according to the way they operate. The main three families of machine learning models are

- supervised learning(Linear Regression, Non-Linear Regression, etc.)
- unsupervised learning(K-Means Clustering, Principal Component Analysis, etc.), and
- Reinforcement learning(Monte Carlo, Q-learning, etc.)

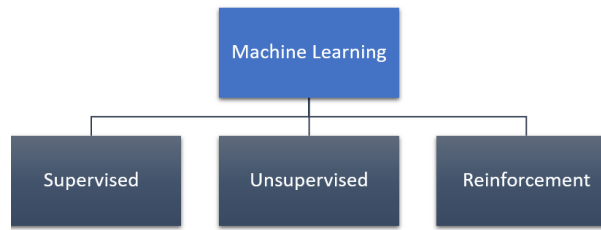


Figure 1.2: Types of ML model

1.4.1 Supervised Learning

When we train a machine using data that is well labeled, we call this process supervised learning, Which means some data is already given with the correct answer. Then the machine is provided with a new set of data, that is, a test data-set, so that the supervised learning algorithm analyses the training data (set of training examples) and produces a correct outcome from labeled data.

Example: Supervised learning is used in Speech Recognition, Bio-Informatics, Object-Recognition for Vision etc. Those were some of the sectors where Supervised Learning has shined and shown its grit in the real world of today.

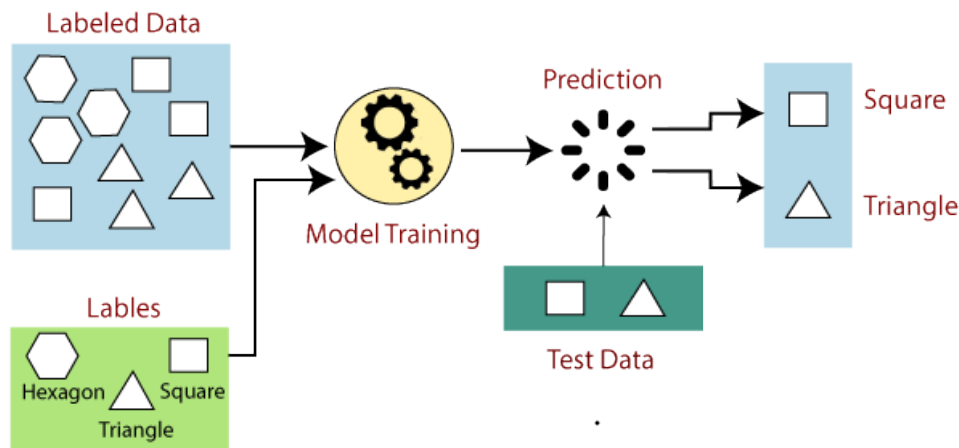


Figure 1.3: Intuition of Supervised Learning Model[7]

In this figure 1.3, some labeled data are given, where labels are hexagon, square and triangle. Then we teach the model which label is hexagon or triangle or square. After that, we provide the model with a test data-set to check how well the model is predicting.

Supervised learning is classified into two categories of algorithms:

Classification: A classification model tries to deduce some conclusion from observed values. A classification problem is when the output variable is a category, such as “Red” or “blue” or “disease” and “no disease”[8].

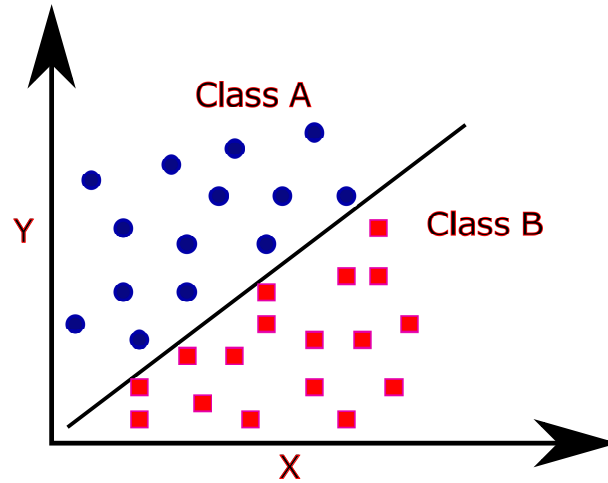


Figure 1.4: Classification

In this figure 1.4, the data-set is classified into two categories. The blue circles are classified as class A and red squares are classified as class B.

Regression: A regression problem is when the output variable is a real value, such as “dollars” or “weight”. There are different types of model that are used as regression but the simplest is the linear regression. It tries to fit the best hyperplane which goes through the training points[8].

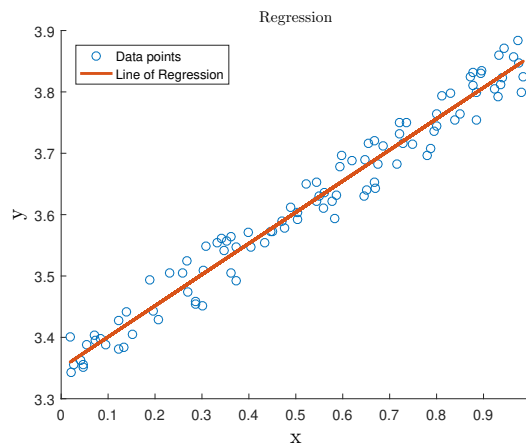


Figure 1.5: Regression

In this figure 1.5, the values on the x-axis are real values, which could be house size or monthly income, etc and the values on the y-axis are real values as well, which could be house price or monthly expenditure, etc. And the relationship between x-values and y-values are linear, that is, their trend is best approximated by a straight line.

1.4.2 Unsupervised Learning

Unsupervised learning is the training of a machine using information that is neither classified nor labeled and allowing the algorithm to act on that information without guidance. Here the task of the machine is to group unsorted information according to

similarities, patterns, and differences without any prior training of data.

Unlike supervised learning, no training is given to the machine. Therefore the machine is restricted to find the hidden structure in unlabeled data by itself.

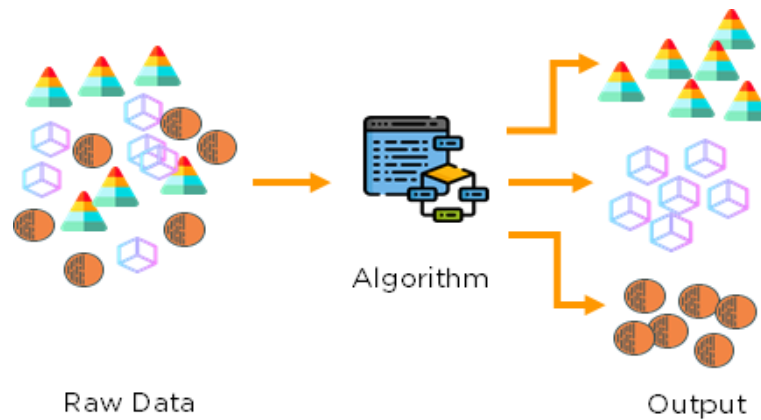


Figure 1.6: Unsupervised Learning[1]

In this figure 1.6, some unlabeled data are given but there are algorithms like k-means clustering, principal component analysis, etc which can categorize these unlabeled data into different categories

Classification of Unsupervised Learning

Clustering: Clustering mainly deals with finding a structure or pattern in a collection of uncategorized data. Unsupervised Clustering algorithms process the data and find natural clusters(groups), if exist.

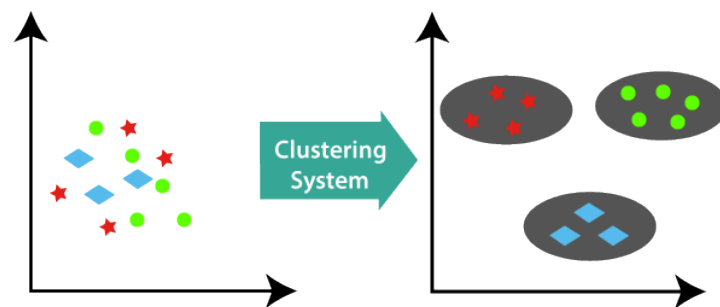


Figure 1.7: An example of Clustering[5]

In this figure 1.7, a data-set is given and there are 3 different shapes but they are all mixed up. So the clustering algorithms work in such a way so that the data set is grouped into different groups with the greatest amount of similarity.

Association: Association rules allow to establish associations amongst data objects inside large databases. This unsupervised technique is about discovering interesting relationships between variables in large databases. For example, people that buy a new home most likely to buy new furniture.

Reinforcement learning: Reinforcement learning (RL) is an area of machine learning concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward[9]. RL is also called feedback-based machine learning algorithm where the agent gets a reward or positive feedback for each good action and a penalty or negative feedback for each bad action.

1.5 Linear Regression

Linear regression is a linear model, e.g. a model that assumes a linear relationship between the input variables (x) and the single output variable (y). More specifically, that y can be calculated from a linear combination of the input variables (x).

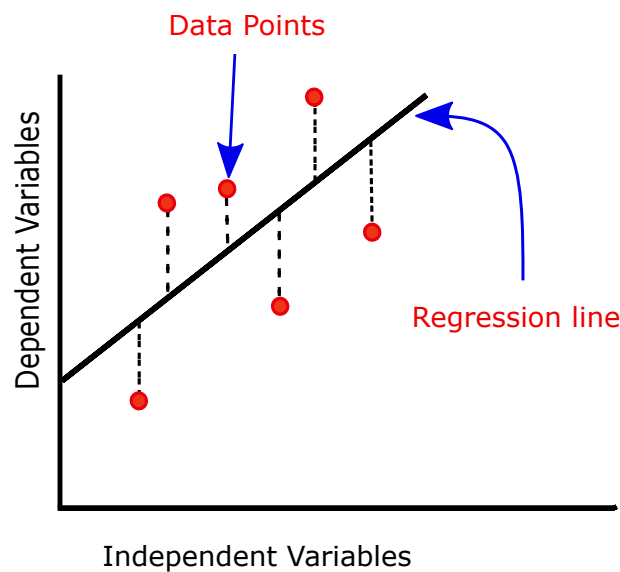


Figure 1.8: Intuition of Linear Regression model

In this figure 1.8, some data points are given and the relationship or trend between the dependent variable and the independent variable is best approximated by a straight line and that straight line is called the “Regression Line”.

We can classify Linear Regression into Simple and Multiple Linear Regression.

1.5.1 Simple Linear Regression

Simple linear regression is used to assess the relationship between one independent variable and one dependent variable. In a Simple Linear Regression problem (a single x and a single y), the form of the model would be:

$$y = \beta_0 + \beta_1 x_1$$

where,

x = the independent variable's value

y = the predicted or dependent variable's value

β_0 = the y -intercept (value of y when all other parameters are set to 0)

β_1 = Slope of the regression line

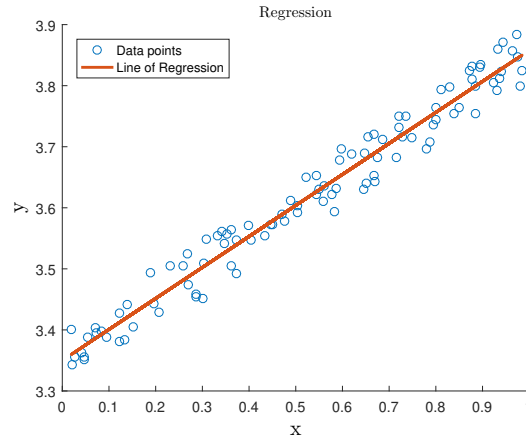


Figure 1.9: Simple Linear Regression model

In this figure 1.9, there is only one independent variable, that is, x and only one dependent variable, that is, y . And Simple Linear Regression algorithm is the best way to represent their trend or relationship.

1.5.2 Multiple Linear regression

Multiple linear regression uses two or more independent variables to predict the outcome of a dependent variable. In a Multiple Linear regression problem (multiple x 's and a single y), the form of the model would be:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where,

y = the predicted value of the dependent variable

β_0 = the y -intercept (value of y when all other parameters are set to 0)

β_n = the regression coefficient of the n th independent variable

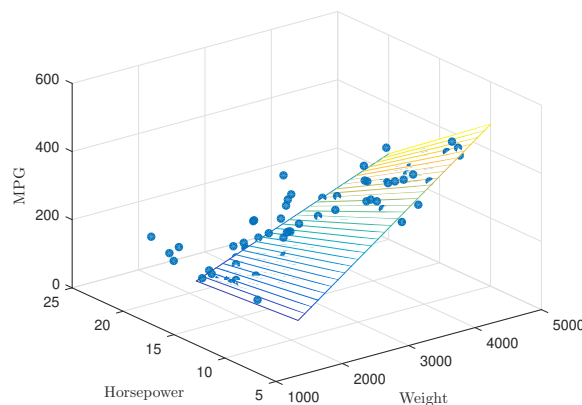


Figure 1.10: Multiple Linear Regression model

In this figure 1.10, there are two independent variables: Horsepower and Weight and one dependent variable: MPG. It is seen that, if Horsepower and Weight increase, then MPG increases as well, that is, there is a linear relationship between dependent variable and

independent variables. And that linear relationship is best approximated by a flat plane or a Multiple Linear Regression line.

1.6 Non-Linear Regression model

Nonlinear regression is a form of regression analysis in which observational data are modeled by a function which is a nonlinear combination of the model parameters and depends on one or more independent variables. The data are fitted by a method of successive approximations.

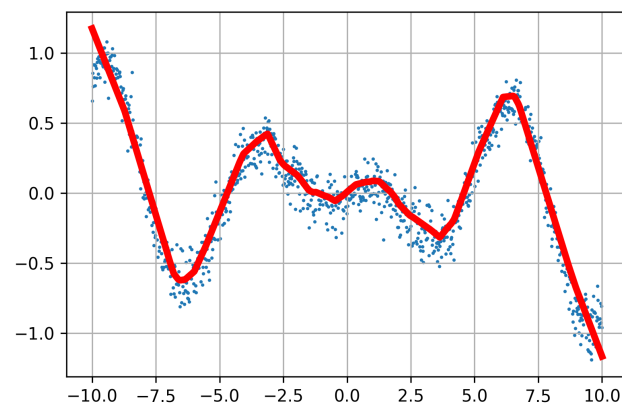


Figure 1.11: A Non-Linear Regression line

In this figure 1.11, there is a independent and dependent variable but the relationships between them is not linear. So the trend between variables are best approximated by a Non-Linear Regression model.

Chapter 2

Linear Regression

In this chapter, we are going to define hypothesis function and cost function of a linear regression model, we'll visualize the cost function in 3D space and to estimate the parameters of hypothesis function we'll use direct method and gradient descent method, we'll discuss how gradient descent algorithm behaves for small and large learning rate α and we'll discuss whether the slope is positive or negative how the gradient descent algorithm always converges to minimum β_j , then we'll discuss about the accuracy and relative error of the linear regression model and in the end we'll give some real life examples which follow linear regression model.

2.1 Introduction

Linear regression is a machine learning algorithm that is used to predict a variable based on one or more independent variables. It predicts the dependent variable as a linear combination of independent variables.

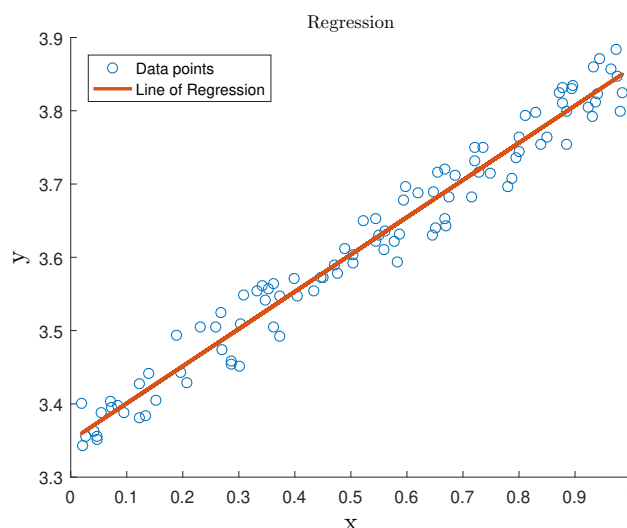


Figure 2.1: Linear Regression

In figure 2.1, we are predicting dependent variable y based on the independent variable x by a linear regression line.

2.2 Model Representation

Let us consider, we want to predict “House-price” based on a feature “House-size”, and we have a collection of sample data, let’s say $(x_1, y_1), (x_1, y_2) \dots (x_n, y_n)$. Where x_i ’s are “House-size” and y_i ’s are “House-price”.

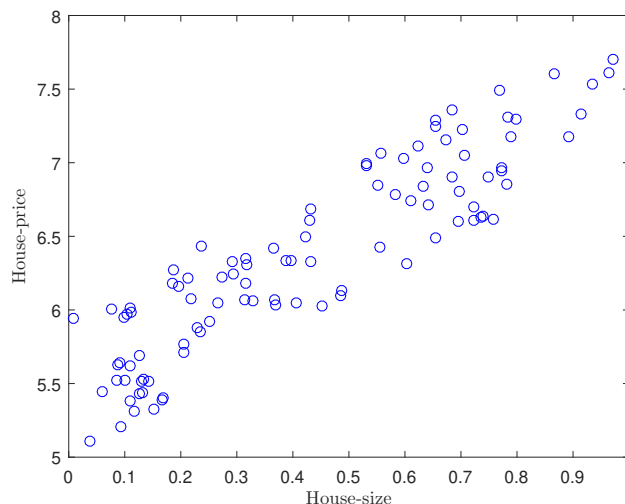


Figure 2.2: *House-size* vs *House-price*

From the visualization of the training data, we see that, the data-set is following a trend, i.e, as “House-size” increases, the “House-price” increases. This trends can be best captured by the following straight line:

$$h_{\beta}(x) = \beta_0 + \beta_1 * x \quad (2.1)$$

$h_{\beta}(x)$ is called hypothesis function, where β_0, β_1 are parameters to be determined. So, Equation 2.1 is the model which best approximates the training data-set.

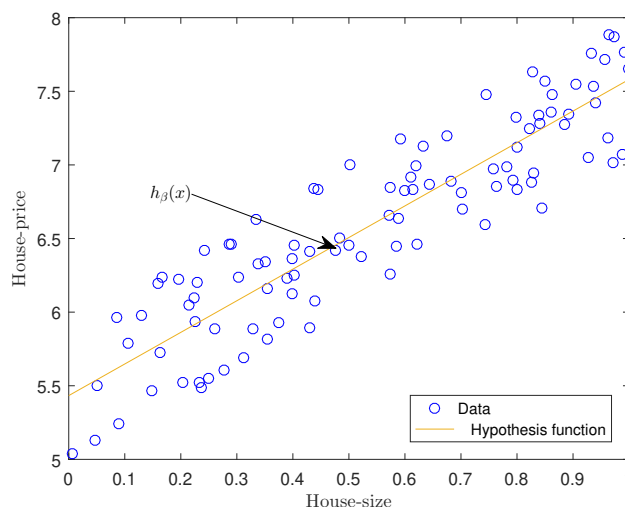


Figure 2.3: The hypothesis function

2.3 Cost function

Cost function is the error between predicted values(the values we get using $h_\beta(x)$) and actual values (y_i), represented as a function. Cost function also measures how well a machine learning model performs for a test data-set.

We denote cost function by $J(\beta_0, \beta_1)$ and is defined as

$$J(\beta_0, \beta_1) = ||y_i - h_\beta(x_i)||_2 \quad (2.2)$$

where,

y_i = exact values

$h_\beta(x)$ = predicted values

$||\cdots||_2 = \ell_2$ norm

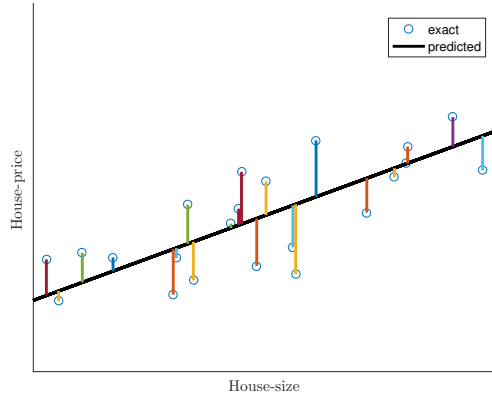


Figure 2.4: Errors between predicted value and exact value

In this figure 2.4, the black color straight line is the regression line and values on that line are predicted values whereas the values denoted by circles are exact values. The distance or error between each predicted and exact value corresponding to a x value is shown by a smaller straight line.

Equation 2.2 can also be written as

$$J(\beta_0, \beta_1) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - h_\beta(x_i))^2} \quad (2.3)$$

From the above equation, we can also conclude that cost function of linear regression is root mean squared error(when ℓ_2 norm is used).

Putting the value of $h_\beta(x)$ in equation 2.3 we get,

$$J(\beta_0, \beta_1) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 * x_i))^2} \quad (2.4)$$

Now, If we visualize the cost function in 3d space, we will see that it takes a paraboloid shape. In figure 2.5, on x-axis and y-axis, we put β_0 and β_1 respectively. And, along z-axis, we put the values from cost function.

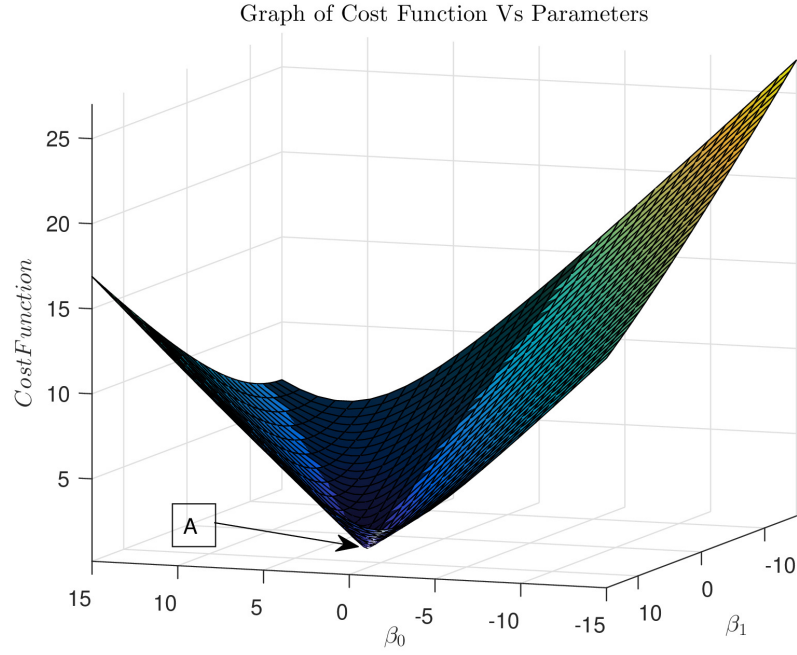


Figure 2.5: 3D visualization of cost function

Let, $\beta_0 = 0$, and $\beta_1 = \beta$ in equation 2.4

$$J(\beta) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \beta * x_i)^2} \quad (2.5)$$

It is to be noted that, when we consider $\beta_0 = 0$, it shifts the data-set to the origin(center) and the mean of the shifted data-set becomes 0(zero). Then it is easier to find the value of other parameter(β_1) and finally we will use β_1 to find β_0 while using direct method to find the values of the parameters.

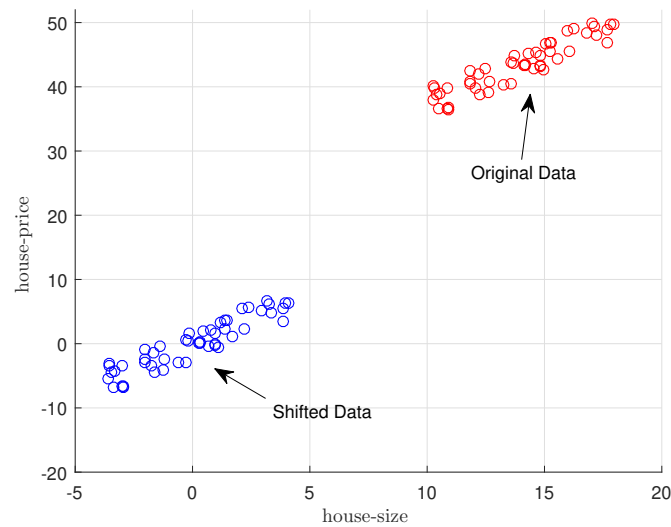


Figure 2.6: Original data-set and shifted data-set

2.4 Minimizing cost function

Previously we have assumed a hypothesis function in section 2.1 and defined cost function for the training data-set. Now, we want our hypothesis function such that it approximates the data-set best and obtain a minimum value for the cost function.

So, Now our goal is to minimize the cost function with respect to the parameters.

$$\text{Goal: } \min_{\beta_0, \beta_1} J(\beta_0, \beta_1)$$

The cost function has minimum at point A in the figure 2.5 in the section 2.3. We can get the minimum value of parameters in two ways: using direct method, gradient descent method, etc.

2.4.1 Direct method

Let us write hypothesis function, cost function and goal as follows:

hypothesis function: $h_\beta(x) = \beta_0 + \beta_1 * x$

cost function: $J(\beta_0, \beta_1) = ||y_i - h_\beta(x_i)||_2$

Goal: $\min_{\beta_0, \beta_1} J(\beta_0, \beta_1)$

Let, $\beta_0 = 0$, $\beta_1 = \beta$. Now we just have to minimize the cost function with respect to β

Goal: $\min_{\beta} J(\beta)$

To minimize the cost function

$$\begin{aligned}\frac{\partial J}{\partial \beta} &= 0 \\ \frac{\partial}{\partial \beta} (||Y - h_\beta(X)||_2^2) &= 0 \\ \frac{\partial}{\partial \beta} [(Y - \beta * X)^T (Y - \beta * X)] &= 0 \\ \frac{\partial}{\partial \beta} (Y^T Y - Y^T \beta X - X^T \beta^T Y + X^T \beta^T \beta X) &= 0 \\ (-Y^T X - X^T Y + 2\beta X^T X) &= 0 \\ \beta &= (X^T X)^{-1} (X^T Y) = \frac{S_{XY}}{S_{XX}}\end{aligned}$$

where, S_{XY} represents the covariance between X and Y and S_{XX} represents the covariance between X and X , that is, the variance of X .

when $\beta_0 \neq 0$

$$y = \beta_0 + \beta_1 x \tag{2.6}$$

Applying average operation in equation 2.6

$$\begin{aligned}\frac{1}{m} \sum_{i=1}^m y &= \frac{1}{m} \sum_{i=1}^m [\beta_0 + \beta_1 x] \\ \bar{y} &= \beta_0 + \beta_1 \bar{x} \\ \beta_0 &= \bar{y} - \beta_1 \bar{x} \\ \beta_0 &= \bar{y} - \frac{S_{XY}}{S_{XX}} \bar{x}\end{aligned}$$

So, finally we get $\beta = \beta_1 = \frac{S_{XY}}{S_{XX}}$ and $\beta_0 = \bar{y} - \frac{S_{XY}}{S_{XX}} \bar{x}$

2.4.2 Gradient descent

Gradient descent is an algorithm that finds optimum of a function. Our goal is to minimize the cost function as much as possible in order to get the best fit line. To minimize the cost function, the model initially selects β_0 and β_1 randomly and then iteratively update these value in order to minimize the cost function.

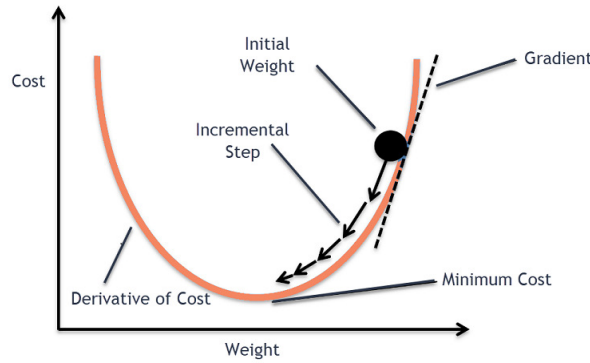


Figure 2.7: Graphical visualization of how gradient descent works

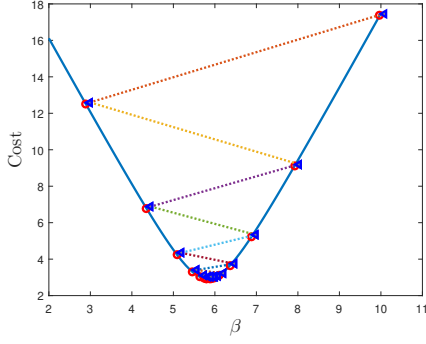
In this figure 2.7, at first we are computing the slope (gradient) of the cost function, a first order derivative, at the current point. And then, moving in the opposite direction of the slope increase from the current point by the computed amount.

Here is the mathematical representation of the algorithm:

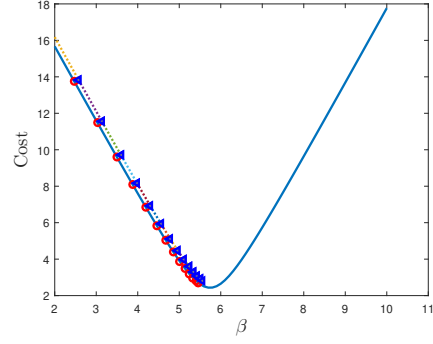
And the algorithm is, Repeat until converge

$$\beta_i := \beta_i - \alpha \frac{\partial}{\partial \beta_i} (J(\beta_0, \beta_1)); i = 0, 1$$

where α is learning rate or step. Depending on the learning rate the gradient descent ($\frac{\partial J}{\partial \beta}$) moves during gradient descent. If α is too high then the gradient moves instably but when α is too low, the algorithm takes more time to reach the minimum point. If $\alpha = 0$, that means the gradient is not moving which means the model isn't learning anything from the gradients.



(a) Large learning rate



(b) Small learning rate

In these two previous figures, we see that, when α is large, the algorithm oscillates too much and takes many iterations to reach the minimum. Similarly, if α is very small, it takes a lot of iterations to reach minimum which is time consuming.

So, the selection of correct α is very important since it guarantees that the algorithm reaches at minimum in a reasonable time. Whether the gradient ($\frac{\partial J}{\partial \beta}$) is positive or negative the algorithm always converges to minimum, here is the graphical explanation of how this is possible:

when slope ($\frac{\partial J}{\partial \beta}$) is +ve:

$\beta_j := \beta_j - \alpha(+\text{ve value}) =$ a updated positive β_j which is smaller than previous β_j . Hence β_j decreases towards the minimum, that is, if the partial derivative is positive then the partial derivative term of gradient descent become negative and it would decrease the weights of the linear regression equation which can be seen in the figure 2.9

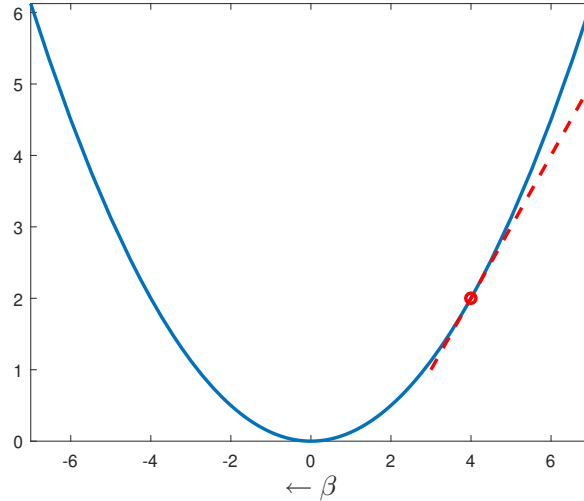


Figure 2.9: β decreases when slope is positive

when slope ($\frac{\partial J}{\partial \beta}$) is -ve:

$\beta_j := \beta_j - \alpha(-\text{ve value}) =$ a updated positive β_j which is larger than the previous β_j . Hence β_j increases towards the minimum, that is, if the partial derivative is negative then the partial derivative term of gradient descent become positive and it would increase the weights of the linear regression equation which can be seen in the figure 2.10

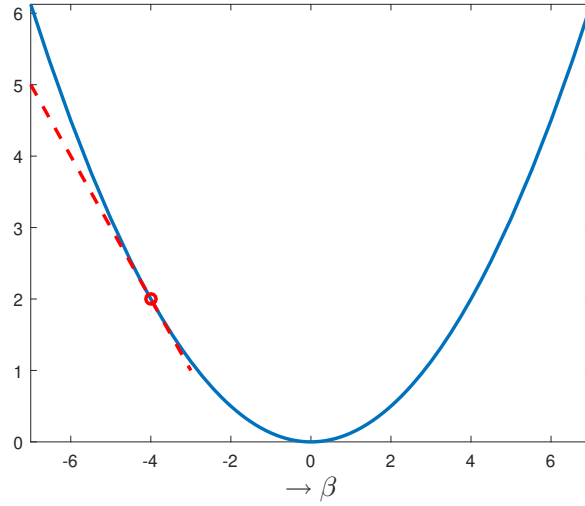


Figure 2.10: β increases when slope is negative

2.5 Accuracy and Relative error

So far, we have built our regression model which best fits the training data-set. Now, we want to check how well the model will predict the test data-set.

$$\text{Model: } \tilde{y} = \beta_0 + \beta_1 x.$$

Let, $\beta_0 = 0$ and $\beta_1 = \beta$. So we have, $\tilde{y} = \beta X$.

$$\text{where, } \beta = \frac{X^T Y}{X^T X} = \frac{\langle X, Y \rangle}{\langle X, X \rangle}$$

We know,

$$\begin{aligned} \text{Cost function} &= \|y - \tilde{y}\|_2^2 \\ &= (y - \tilde{y})^T (y - \tilde{y}) \\ &= (y - \beta x)^T (y - \beta x) \\ &= \left(y - \frac{\langle X, Y \rangle}{\langle X, X \rangle} x\right)^T \left(y - \frac{\langle X, Y \rangle}{\langle X, X \rangle} x\right) \\ &= Y^T Y - Y^T \frac{\langle X, Y \rangle}{\langle X, X \rangle} X - \frac{\langle X, Y \rangle}{\langle X, X \rangle} X^T Y + \frac{\langle X, Y \rangle \langle X, Y \rangle}{\langle X, X \rangle \langle X, X \rangle} X^T X \\ &= Y^T Y - 2 \frac{\langle X, Y \rangle}{\langle X, X \rangle} X^T Y + \frac{\langle X, Y \rangle \langle X, Y \rangle}{\langle X, X \rangle} \\ &= Y^T Y - 2 \frac{\langle X, Y \rangle \langle X, Y \rangle}{\langle X, X \rangle} + \frac{\langle X, Y \rangle \langle X, Y \rangle}{\langle X, X \rangle} \\ &= Y^T Y - \frac{\langle X, Y \rangle \langle X, Y \rangle}{\langle X, X \rangle} \\ &= Y^T Y \left(1 - \frac{\langle X, Y \rangle \langle X, Y \rangle}{\langle X, X \rangle \langle Y, Y \rangle}\right) \\ &= R_{YY} (1 - \rho_{XY}^2) \end{aligned}$$

where, R_{xy} and ρ_{xy} are Pearson's correlation coefficients.

Relative error of the model is : $= \frac{\|y - \tilde{y}\|_2^2}{\|y\|_2^2} = \frac{R_{xy}(1 - \rho_{xy}^2)}{R_{xy}} = 1 - \rho_{xy}^2$

For example, if $\rho_{xy} = 0.8$ (i.e, x 's and y 's are strongly correlated)). Then, the **relative error** of the model is $= 1 - (0.8)^2 = 0.36 = 36\%$

Which means that, two data-sets are strongly correlated and the model estimates the data-set with 64% accuracy.

2.6 Linear Regression examples

Example-1

A engine making company might want to get the relation between Horsepower and Acceleration. They might fit a simple linear regression model using Horsepower as the predictor variable and Acceleration as the response variable. The regression model would take the following form

$$\text{Acceleration} = \beta_0 + \beta_1 * (\text{Horsepower}) \quad (2.7)$$

Let's consider a sample data-set given below as a table:

Acceleration	12	11.50	11	12	10.5	10	9	8.5	10	...
Horsepower	130	165	150	150	140	198	220	215	225	...

Now, if we plot "Acceleration" vs "Horsepower" using the data-set, then we get the following graph:

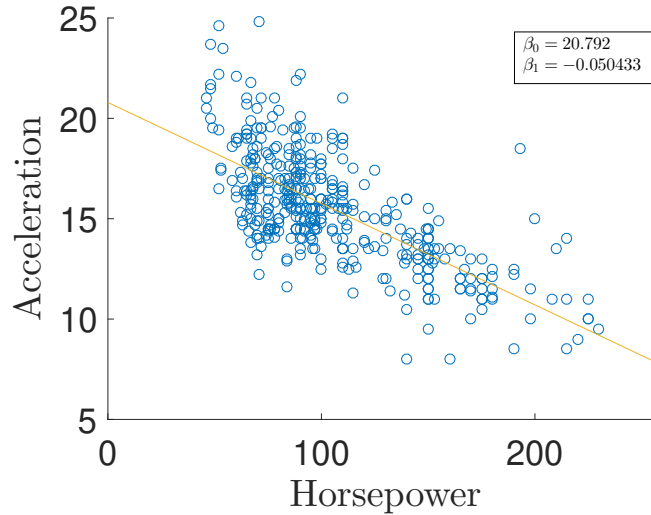


Figure 2.11: A linear regression example

Since "Acceleration" and "Horsepower" are linearly related, applying the Normal equation method or gradient descent method we get $\beta_0 = 20.792$ and $\beta_1 = -0.054033$. So equation 2.7 becomes:

$$\text{Acceleration} = 20.792 - 0.054033 * (\text{Horsepower}) \quad (2.8)$$

If a car engine company wants to approximate the Acceleration of a car with Horsepower of 250, then from the equation 2.8 the approximated acceleration is

$$\begin{aligned}\text{Acceleration} &= 20.792 - 0.054033 * (250) \\ \text{Acceleration} &= 7.28\end{aligned}$$

Example-2

A botanist might want to get the relation between sepal width and sepal length. They might fit a simple linear regression model using sepal width as the predictor variable and sepal length as the response variable. The regression model would take the following form

$$\text{sepal length} = \beta_0 + \beta_1 * (\text{sepal width}) \quad (2.9)$$

Let's consider a sample data-set given below as a table:

sepal length	3.50	3	3.20	3.10	3.60	3.90	3.40	3.40	2.90	...
sepal width	5.10	4.90	4.70	4.60	5	5.40	4.60	5	4.40	...

Now, if we plot “sepal length” vs “sepal width” using the data-set, then we get the following graph:

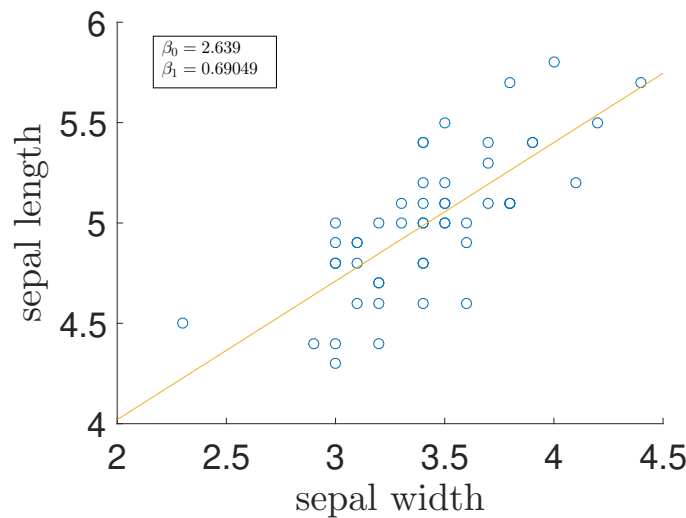


Figure 2.12: A linear regression example

Since “sepal length” and “sepal width” are linearly related, applying the Normal equation method or gradient descent method we get $\beta_0 = 2.639$ and $\beta_1 = 0.69049$. So equation 2.9 becomes:

$$\text{sepal length} = 2.639 + 0.69049 * (\text{sepal width}) \quad (2.10)$$

If a botanist wants to approximate the sepal length of a flower with 4.2 sepal length, then from the equation 2.10 the approximated sepal length is

$$\begin{aligned}\text{sepal length} &= 2.639 + 0.69049 * (4.2) \\ \text{sepal length} &= 5.539058\end{aligned}$$

Example-3

An online-based startup might want to get the relation between Length Of Membership and Yearly Amount Spent. They might fit a simple linear regression model using Length Of Membership as the predictor variable and Yearly Amount Spent as the response variable. The regression model would take the following form

$$\text{Yearly Amount Spent} = \beta_0 + \beta_1 * (\text{Length Of Membership}) \quad (2.11)$$

Let's consider a sample data-set given below as a table:

Length of Membership	4.08	2.66	4.10	3.12	4.44	5.49	4.68	...
Yearly Amount Spent	587.95	392.20	487.54	581.85	599.40	637.10	521.57	...

Now, if we plot “Length of Membership” vs “Yearly Amount Spent” using the data-set, then we get the following graph:

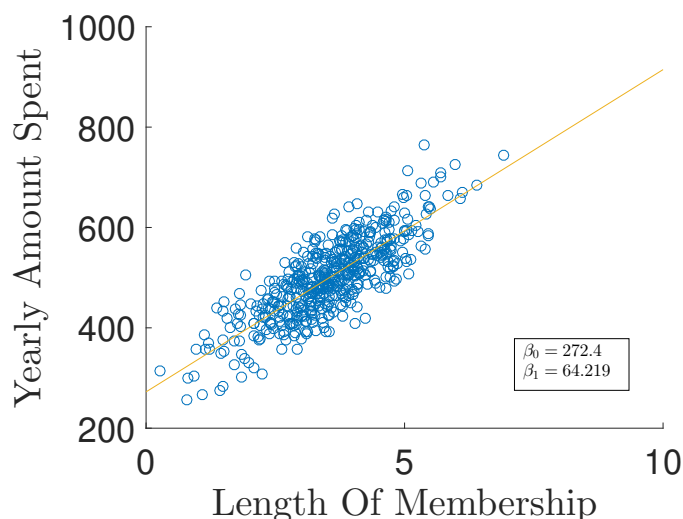


Figure 2.13: A linear regression example

Since “Yearly Amount Spent” and “Length of Membership” are linearly related, applying the Normal equation method or gradient descent method we get $\beta_0 = 272.4$ and $\beta_1 = 64.219$. So equation 2.10 becomes:

$$\text{Yearly Amount Spent} = 272.4 + 64.219 * (\text{Length of Membership}) \quad (2.12)$$

If the startup wants to predict the Yearly Amount Spent of a customer with 8 years Length Of Membership, then from the equation 2.12 the predicted Yearly Amount Spent is

$$\begin{aligned} \text{Yearly Amount Spent} &= 272.4 + 64.219 * (8) \\ \text{Yearly Amount Spent} &= 786.15 \end{aligned}$$

Chapter 3

Logistic Regression

In this chapter, we will give an introduction of Logistic Regression and briefly discuss about different types of Logistic Regression and then we will show what are the differences between Linear and Logistic Regression model using graphs and we will describe why Linear Regression is not a good fit for classification problem and then we will define hypothesis function and cost function for Logistic Regression and then we will show what will happen if we put hypothesis function of Logistic Regression in Linear Regression cost function, then we will discuss about linear and non-linear decision boundary with some examples and in the end we will present some applications of Logistic Regression.

3.1 Introduction

One of the most widely used Machine Learning algorithms, under the category of Supervised Learning, is logistic regression. It is a classification algorithm used to predict the probability of a target or dependent variable. The value of the target or dependent variable is either 1 (stands for yes/success) or 0 (stands for no/failure). Email spam or not spam, online transaction fraud or not fraud, and tumor malignant or benign are a few examples of classification problems. The logistic sigmoid function is used by logistic regression to convert its output and produce a probability value. In the graph of this function we set a hypothetical threshold (usually 0.5) above which we predict the result to be in class 1 and below which we predict the result to be in class 2.

Apart from how they are applied, logistic regression and linear regression are very similar. Regression problems are resolved using linear regression and classification problems will be solved using logistic regression. In logistic regression, we fit a “S” shaped logistic function, which predicts two extreme values (0 or 1), rather than a regression line.

The curve of the logistic function in the next figure shows the probability of events such as whether or not the cells are malignant, whether or not a mouse is obese depending on its weight, and so on.

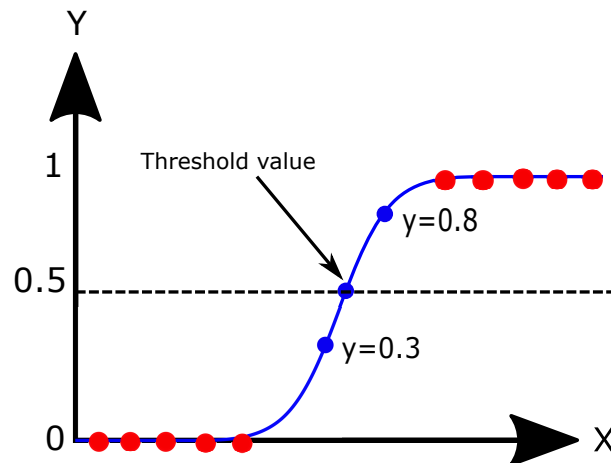


Figure 3.1: Logistic Regression

3.2 Types of Logistic Regression

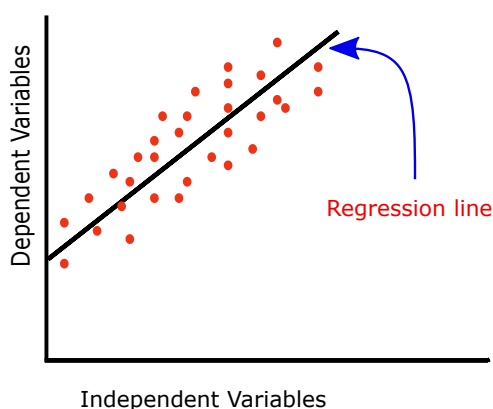
Three different types of logistic regression can be classified depending on the categories:

Binomial: In binomial logistic regression, there are only two possible values of the target or dependent variable. Example: 0 or 1, pass or fail, etc.

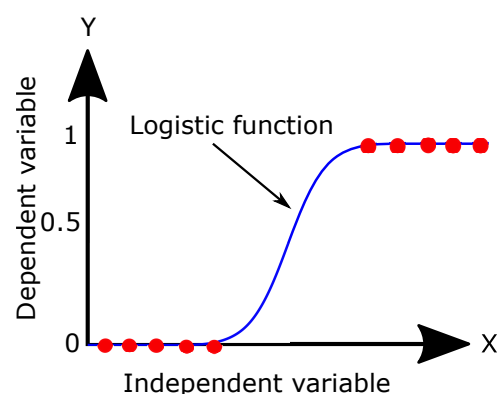
Multinomial: In multinomial logistic regression, the dependent variable may be three or more possible **unordered** types, such as “cat,” “dogs,” or “sheep”.

Ordinal: In ordinal logistic regression, the dependent variable may be three or more possible **ordered** types, such as “low”, “Medium”, or “High”.

3.3 Differences between Linear and Logistic Regression



(a) Linear Regression



(b) Logistic Regression

Logistic Regression and Linear Regression are sort of similar except that how they are used in a model. Linear Regression is used for solving regression problems whereas Logistic Regression is used for solving classification problem. In Linear Regression we fit a straight line in the data but in case of Logistic Regression we fit a “S” shaped logistic

function. In Linear Regression, the independent and dependent variable must have a linear relationship but in Logistic Regression it is not required. In Linear Regression, output values are continuous values but in Logistic Regression the output values are discrete values.

3.4 Why Linear Regression is not a good fit for classification problem?

Let us consider, we have a data-set where tumor size is independent variable and dependent variable can take either 0 or 1 and where 0 represents the tumor is benign and 1 represents the tumor is malignant. The visual representation of this data-set is given in Figure 3.3.

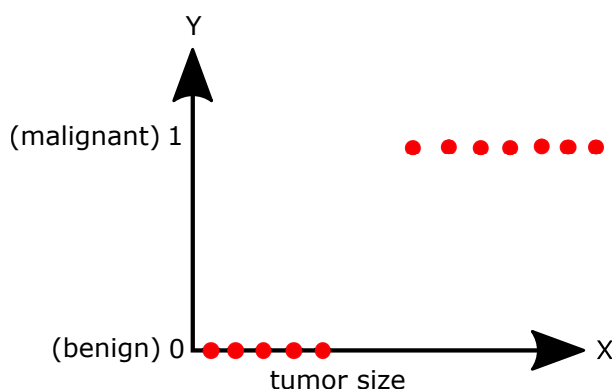
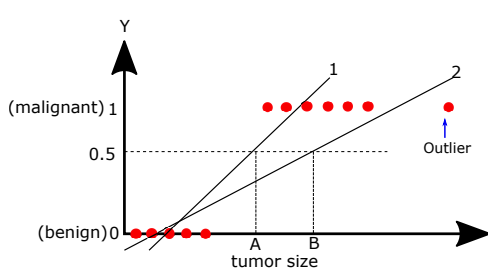
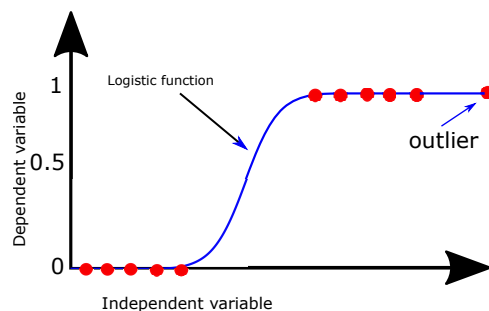


Figure 3.3: Visualization of a data-set of classification problem

Now, we apply the algorithm that we already know, that is, linear regression, so we fit a straight line to the training data and we see that the system works well with the straight line 1. At first, we set a threshold which is 0.5, if the outcome is less than 0.5 then the tumor is benign and otherwise it is malignant. Corresponding to this threshold point we get a critical point A on the x-axis, the tumor size less than that critical point means it is benign and otherwise it is malignant. So far, the straight line 1 is predicting the training data quite well but if we add new valid but outlier data in the training data-set then the Linear Regression line is changed to 2 and that changes how it distinguishes the borderline cases. For straight line 2 the critical point is now B, everything to the right of this point we predict as 1 and everything to the left of that point we predict as 0 and this seems a pretty bad prediction for linear regression to have done because mathematically it is showing that all the points left to B are benign but in reality there are still few points which are actually malignant. All of these problems can be solved if we use a “S” shaped curve that is a sigmoid function. It can even catch all the outliers as well. Further, regression line is not a best fit because it is likely to predict “ > 1 ” and “ < 0 ” class probabilities which are not true for classification problem.



(a) Visualization of why linear regression is not a good fit for classification problem



(b) Logistic Regression

3.5 Model representation

For linear regression the hypothesis function is:

$$h_{\Theta}(x) = \Theta^T x$$

On the other hand, the hypothesis function of logistic regression is more generalization of linear regression hypothesis function given that it makes use of sigmoid function:

$$h_{\Theta}(x) = g(\Theta^T x)$$

where,

$$g(z) = \frac{1}{1 + e^{-z}} \quad (3.1)$$

Finally the hypothesis function for logistic regression is :

$$h_{\Theta}(x) = \frac{1}{1 + e^{-\Theta^T x}}$$

The equation 3.1 is the logistic function and is also known as the sigmoid function. A sigmoid function is a real function. This function limits the output to a range between 0 and 1. It is a “S” shaped curve and it has exactly one inflection point[3].

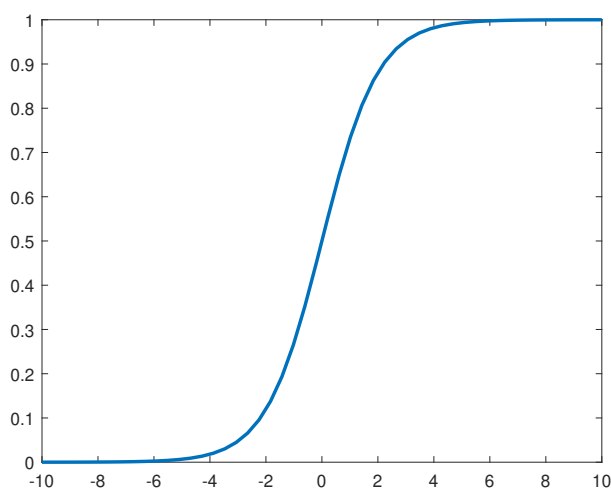


Figure 3.5: A sigmoid function

3.5.1 Threshold value

A threshold value in logistic regression classifies the borderline cases. It is also called decision boundary above which the algorithm predicts 1 and below which 0. Usually that threshold value is 0.5. That is, the decision boundary is $h_{\theta}(x) = 0.5$,

- if $h_{\theta}(x) \geq 0.5$, then predict $y = 1$.
- if $h_{\theta}(x) < 0.5$, then predict $y = 0$.

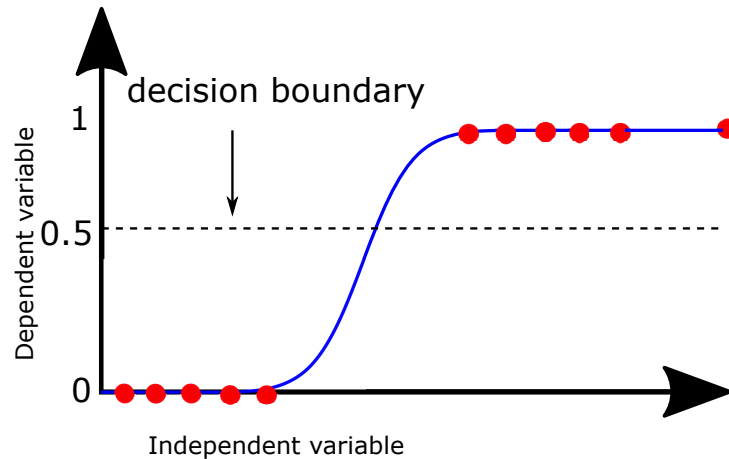


Figure 3.6: Visualization of Decision boundary

3.6 Cost Function

3.6.1 Linear regression cost function and logistic regression hypothesis

For linear regression the cost function is:

$$\text{cost}(h_{\theta}(x), y) = (h_{\theta}(x) - y)^2$$

Now if we plug in the hypothesis function of logistic regression in this cost function of linear regression and if we plot the cost function then we will get a non-convex graph as given below. From this graph finding a global minimum of θ which minimizes the error is not guaranteed while using gradient descent method.

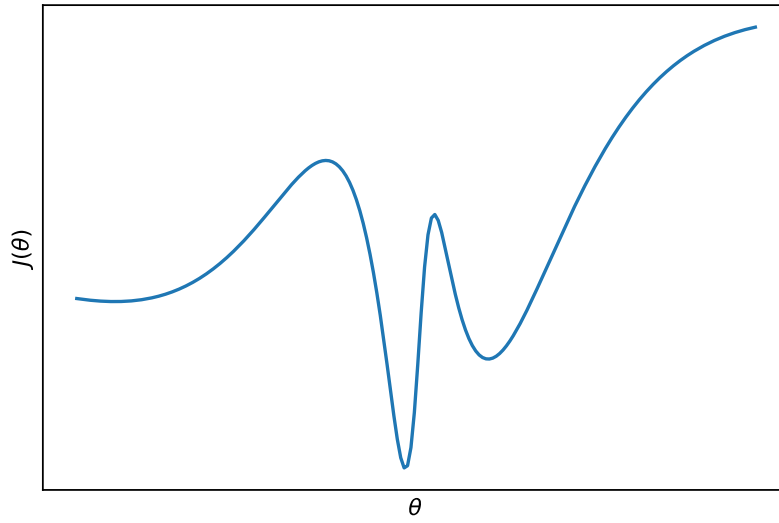


Figure 3.7: Non-convex cost function

So to avoid this situation, we have to come up with a different cost function for logistic regression so that we can get convex function and then algorithm like gradient descent can be applied to minimize the cost function.

3.6.2 Logistic regression cost function

$$J_{\theta}(x) = \begin{cases} -\log(h_{\theta}(x)); & \text{If } y = 1 \\ -\log(1 - h_{\theta}(x)); & \text{If } y = 0 \end{cases}$$

The above function can be written together as :

$$J(\Theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(h_{\Theta}(x(i))) + (1 - y^{(i)}) \log(1 - h_{\Theta}(x(i)))]$$

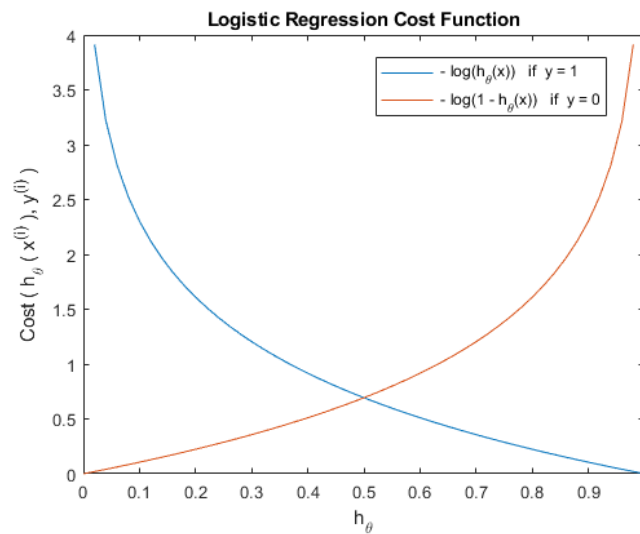


Figure 3.8: Cost function

3.7 Decision boundary

The fundamental task of logistic regression is trying to find out a decision boundary for classification problem, in one side of which the algorithm classifies the data value to a class and other side of which to another class.

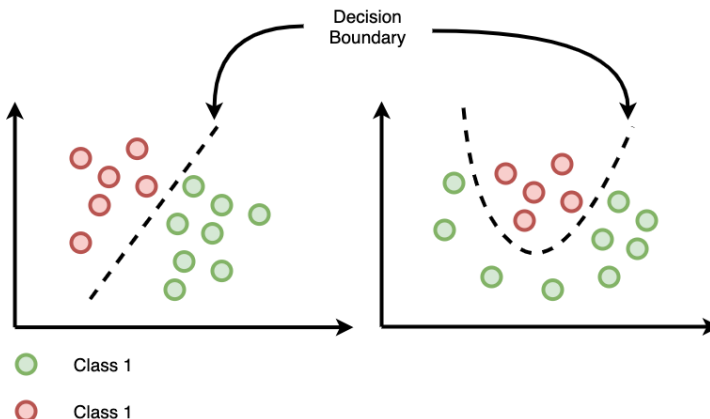


Figure 3.9: Decision boundaries

In above figure, the dashed lines are decision boundary which separates two different classes in two different region. In logistic regression our motivation is to figure out a best fit decision boundary for the training data so that the algorithm will be able to predict which class a new feature set might belong to.

3.7.1 Linear decision boundary

When the decision boundary line(plane for higher dimension) is a linear function of given features then we call it linear decision boundary. If we have D dimensional input space then the decision boundary is a $D-1$ dimensional hyperplane. For example, 2D decision boundaries(planess) for a set of 3 dimensional features and 1D decision boundaries(lines) for a set of 2 dimensional features.

An example of linear decision boundary:

For logistic regression,

$$h_{\Theta}(x) = g(\Theta^T x)$$

where,

$$g(z) = \frac{1}{1 + e^{-z}} \text{ and } z = \theta^T x$$

Suppose,

- predict $y=1$ if $h_{\theta}(x) = g(\theta^T x) \geq 0.5$ whenever $\theta^T x \geq 0$
- predict $y=0$ if $h_{\theta}(x) = g(\theta^T x) < 0.5$ whenever $\theta^T x < 0$

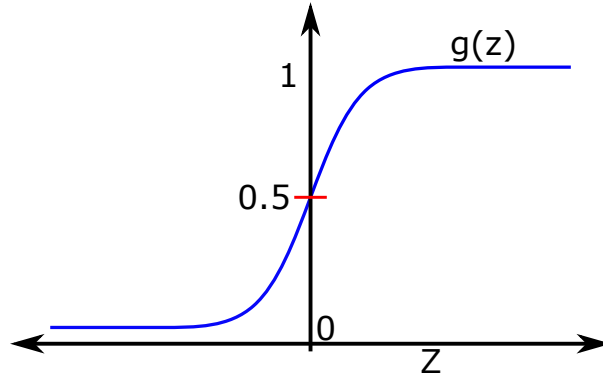


Figure 3.10: Linear decision boundary

To summarize what we just worked out, we saw that if we decide to predict whether $y=0$ or $y=1$ depending on whether the estimated probability (or $h_\theta(x)$) is greater than or equal to 0.5 (threshold point) or whether it's less than 0.5, then that's the same as saying that we'll predict $y=1$ whenever $\theta^T x \geq 0$ and will predict $y=0$ whenever $\theta^T x < 0$.

An example of linear decision boundary

Let's see another example of decision boundary. Suppose the hypothesis function for the data-set shown in graph given below is

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Let, $\theta_0 = -3$, $\theta_1 = 1$, $\theta_2 = 1$

Now, let's try to figure out where our $h_\theta(x)$ will end up predicting $y=1$ and where $y=0$.

Predict $y=1$ if

$$-3 + x_1 + x_2 \geq 0 \tag{3.2}$$

So for any example with features x_1 and x_2 that satisfies the equation 3.2, then the hypothesis will think that $y=1$ is more likely.

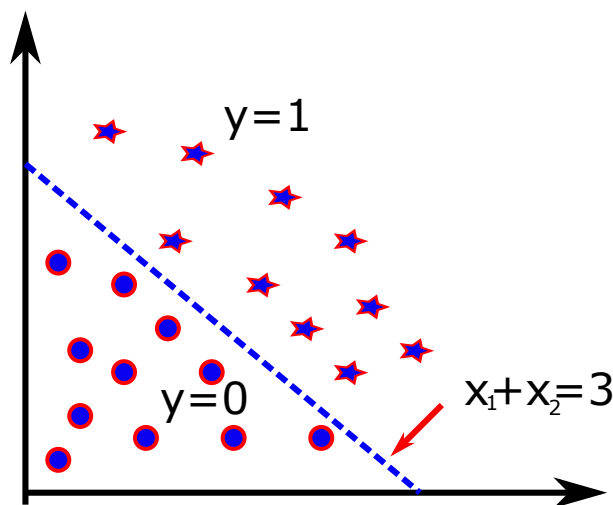


Figure 3.11: Linear decision boundary

3.7.2 Non-linear decision boundary

Unlike linear decision boundary, the non-linear decision boundary is a non-linear function of features.

Let's say we have a classification problem and after plotting the data points we see that a non-linear decision boundary(a circle) is the best fit for the particular problem. Suppose, the hypothesis function for this problem is:

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

Let, $\theta_0 = -1$, $\theta_1 = 0$, $\theta_2 = 0$, $\theta_3 = 1$, $\theta_4 = 1$

Now, let's try to figure out where our $h_{\theta}(x)$ will end up predicting $y=1$ and where $y=0$.

Predict $y=1$ if

$$\begin{aligned} -1 + x_1^2 + x_2^2 &\geq 0 \\ \text{or, } x_1^2 + x_2^2 &\geq 1 \end{aligned}$$

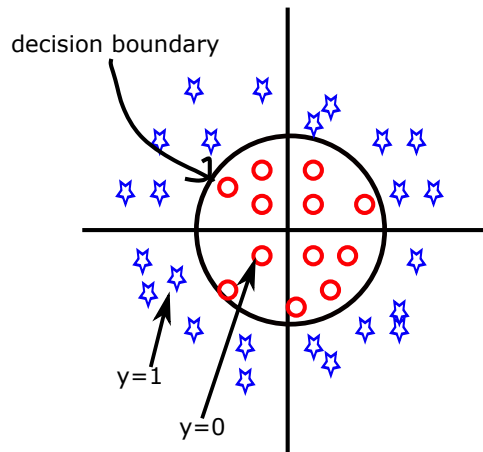


Figure 3.12: Non-Linear decision boundary

3.8 Applications of Logistic Regression

Example-1

A company would like to know whether or not the quantity of words in an email and the origin of the email both increase the likelihood that the email is spam. To determine the nature of the connection that exists between these two factors and the likelihood that an email contains spam, researchers can make use of the technique of logistic regression. The model's response variable will be "spam", with two possible outcomes:

- The email is spam.
- The email is not spam.

The results of the model will inform the company exactly how differences in word count and country of origin affect the likelihood that a specific email is spam. These results will also tell the organization exactly how these variables affect the likelihood that a certain email is spam[2].

Example-2

A credit card firm is interested in conducting research to see whether or not the total amount of a transaction and the customer's credit score have any influence on the possibility that a particular transaction is fraudulent. The organization is able to acquire a better grasp of the connection that exists between these two predictor variables and the possibility that a transaction is fraudulent by employing a statistical technique known as logistic regression. The model's response variable will be "fraudulent" with two possible outcomes:

- The transaction is fraudulent.
- The transaction is not fraudulent.

The findings of the model will notify the organization precisely how variations in transaction amount and credit score affect the probability that a particular transaction is fraudulent. The organization is also able to utilize the fitted logistic regression model to assess the probability that a particular transaction is fraudulent. This capability allows the business to be more proactive in preventing fraudulent activity. This forecast is formed by taking into account the magnitude of the transaction in question in addition to the credit rating of the individual who carried it out.

Chapter 4

Regularization

In this chapter, we will define regularization and will discuss about two most frequently used terms in regularization: bias and variance, we will discuss about over-fitting, under-fitting using graphs and reasons for why they occur and how to avoid them, and then we will show how we can get a good-fit model from an under-fit or over-fit model using ℓ_1 and ℓ_2 regularization and how we can get the value for tuning parameter λ using L-curve method and finally we will discuss about the differences between ℓ_1 and ℓ_2 regularization.

4.1 Introduction

Main objective of learning a new machine learning model is to use that model in a data-set and trying to figure out the relationship between independent and dependent variables and in future to use that model for prediction. When we actually try to fit a model in a data-set, we sometimes end up with such a model which either under-fits the data-set, that is, the model is not able to represent the true relationship between the independent and dependent variables or over-fits the data-set which means the model is trying too hard to capture all the features in the training data set.

When a model under-fits the data-set, we say the model has low variance and high bias. In case of over-fits, the model has high variance and low bias. The regularization techniques do a trade-off between under-fit and over-fit model to get a good-fit model.

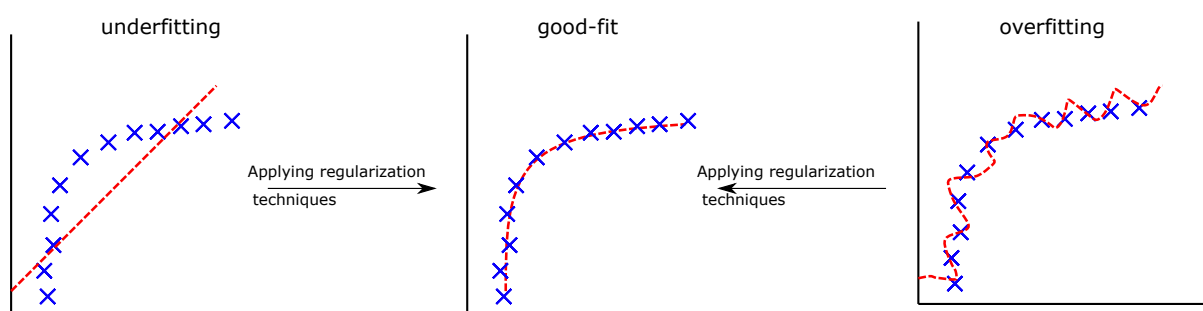


Figure 4.1: Problems of over-fitting

Now we will discuss about what we mean by bias and variance in a model.

- **Bias:** The machine learning model itself experiences bias as a result of very simplistic assumptions made throughout the ML process. It is actually error rate of

the training data when the error rate has a high value, we call it high bias and when the error rate has a low value, we call it low bias. When a model has high bias, it fails to capture the data trends and the model becomes overly simplified[12].

- **Variance:** Variance describes how the model changes when various parts of the training data set are used. Simply said, variance is the model prediction's ability to vary, or how much the ML function may change based on the input data set. Models with many features and a high level of complexity are the source of variation. When a model tries too hard to capture all the features in a data-set then it produces high variance. A model with high variance produces high error across different test data-set[12].

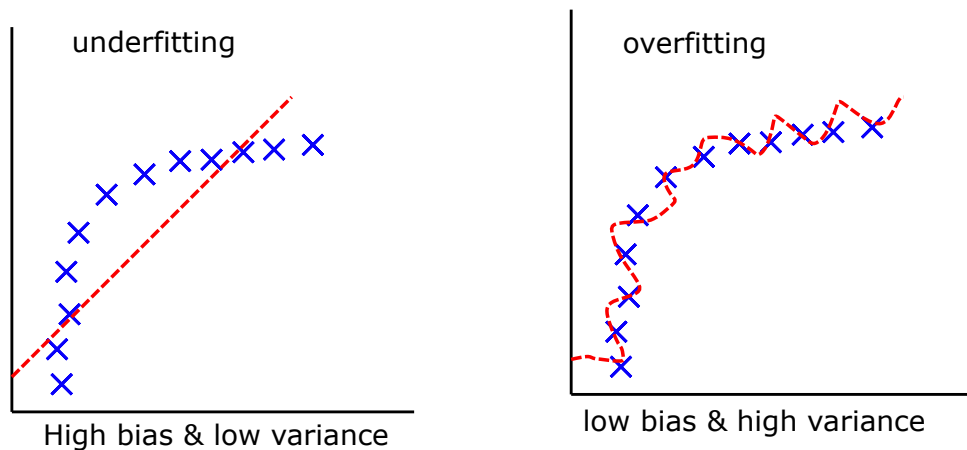


Figure 4.2: Visualization of bias and variance in the model

4.2 Over-fitting

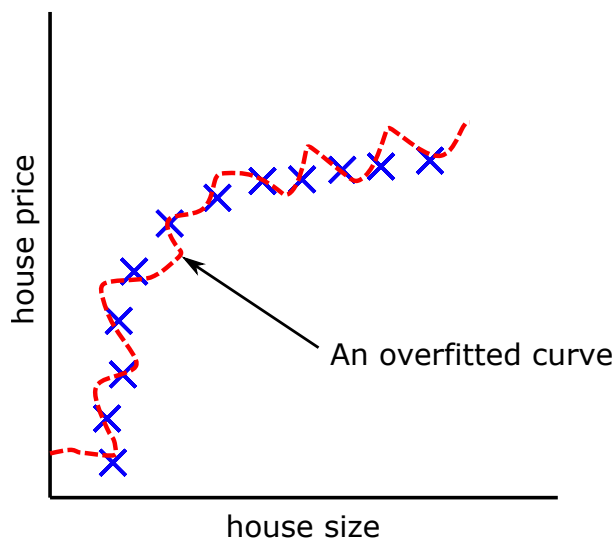


Figure 4.3: An over-fit model

Over-fitting happens when a machine learning model is restricted to the training set and is unable to perform effectively on unseen data or test data. The model rejects every

new data point during prediction because it is not flexible enough to forecast new data points. The over-fit model has a relatively low bias but a high variance.

4.2.1 Reasons for over-fitting

There are multiple reasons which cause a model to over-fit the training data.

- when the model is too complex.
- when the number of data points in training data-set is not enough.
- when data has a lot of noise, that is, there are outliers and errors in training data.

4.2.2 How to avoid over-fitting?

- If the data-set contains redundant features, removing them can prevent a model to over-fit.
- Taking enough data-points in the training data-set.
- Applying regularization techniques such as ℓ_2 regularization and ℓ_1 regularization.

4.3 Under-fitting

When a statistical model or machine learning algorithm is unable to recognize the fundamental pattern in the data, it is said to have under-fitting. To prevent over-fitting, We could terminate the training earlier. However, it could also result in the model not being able to learn enough from the training data, which could make it challenging to identify the main trend. Since the model is not able to learn enough about the trend of data-set in case of under-fitting, it produces less accuracy and results in incorrect predictions.

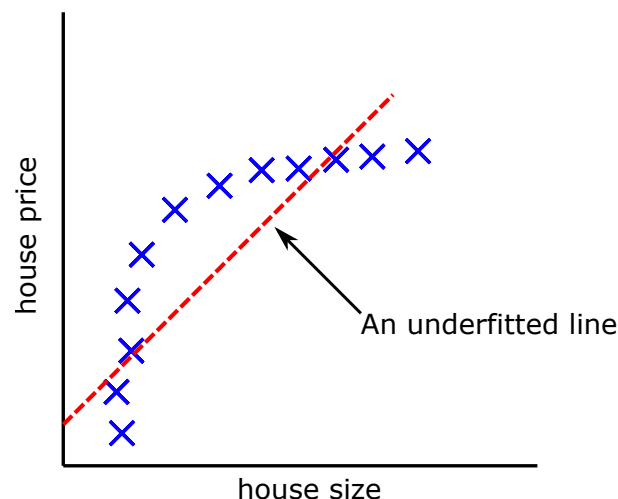


Figure 4.4: An under-fit model

4.3.1 Reasons for under-fitting

- too simple model causes under-fitting.
- when not enough data points are used in the model.
- when the data used for training is not cleaned and has noise in it.

4.3.2 How to avoid under-fitting?

- by adding more data points.
- by Cleaning the data or reducing the noise from the data.
- by increasing model complexity.
- by prolonging the model's training period.

4.4 What is good-fit?

We say a model is a good-fit for the provided data if the model predicts almost same outcome as the true value. And a good-fit model produces almost same error across different data-set. A good-fit model is a model somewhere between a under-fit and over-fit model. A under-fit model has high bias and low-variance and over-fit model has low bias and high variance whereas a good-fit model has moderate amount of bias and variance.

When we train a model, at the very beginning the model is very simple hence a under-fit model. As the model learns over time, it starts to understand the trend in the data and hence the error on the training data-set reduces as well as on the test data-set. If we train the model for too long, the model starts to learn about the noise in the data and the model approaches towards a over-fit model. So to get a good-fit model we have to stop training the model as soon as we observe that the error increases.

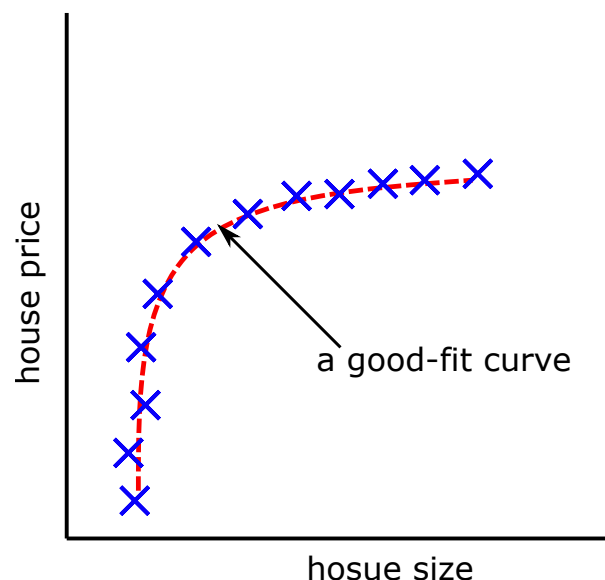


Figure 4.5: A good-fit model

4.5 Addressing over-fitting

let us think, over-fitting is occurring. Now what can we do to address it?

when we had 1D or 2D data-set, then we could easily plot them and easily could figure out which degree of polynomial is “good-fit” or “over-fit” the data-set. but, we might not always have 1D or 2D data set. In fact more often we may have learning problems that where we just have a lot of features and there is not just a matter of selecting what degree of polynomial and in fact it when we have so many features it also becomes much harder to plot the data and becomes much harder to visualize it. To decide what features to keep or not so that we can modify the model from “over-fit” to “good-fit”.

There are 2 main options to address over-fitting:

1. Reduce number of features. We can do this by 2 different ways:

- (a) Manually select which features to keep.
- (b) Model selection algorithm(automatically select which features to through out or keep).

The idea of reducing the number of features can work well and can reduce over-fitting. But the disadvantage is that by throwing away some of the features is also throwing away some of the information we have about the problem.

2. Regularization

- Keep all the features, but reduce magnitude/values of parameters.
- Works well when we have a lot of features, each of which contributes a bit to predicting y .

Here we will only talk about regularization techniques.

4.6 Regularization

Regularization is a general phrase that refers to techniques that need the learning algorithm to construct a simpler model. In actuality, this frequently results in a somewhat higher bias but substantially lower variance. The problem is also known as the bias-variance trade-off.

Previously in section 2.1, we have discussed that, too simple model produces high bias and too complex model produces high variance. But we want such a model which is neither “highly biased” nor have “high variance” but has moderate amount of variance and bias in the model. The regularization techniques help to achieve that kind of model by adding a regularization term in the objective function of the model.

4.6.1 Types of Regularization technique:

There are mainly 2 different types of techniques used in regularization

- **ℓ_2 regularization:** Also known as Ridge Regression. We use ℓ_2 norm in the regularization term.

- **ℓ_1 regularization:** Also known as Lasso Regression. We use ℓ_1 norm in the regularization term.

4.7 ℓ_2 regularization

ℓ_2 regularization adds ℓ_2 norm of coefficient as penalty term or regularization term to the error function or objective function.

Let us consider, we have a problem where we have n numbers of features. The data values corresponding to these features are given by the matrix A and the output is given by the matrix b and the coefficients corresponding to each features is given by the matrix W . So our objective function is:

$$\min ||AW - b||_2^2$$

In this minimization problem on W , we are imposing no restriction, it can be anything if it satisfies the minimization problem. We are assuming that this is a over-fit model, that is, the model have learnt about some features which are not so important in prediction. So, we want to reduce their weight instead of completely vanishing them. So, that's why we now want to impose restriction on W by adding an extra regularization term with the objective function.

$$\min (||AW - b||_2^2 + \lambda ||W||_2^2) \quad (4.1)$$

where, λ is regularization parameter.

Regularization can be conceptualized as an additional penalty term or constraint, as represented in the following picture. The global cost minimum is what we are looking for without regularization. By including a regularization penalty, we change our goal to minimizing the cost function while staying within our “budget” (the gray-shaded ball).

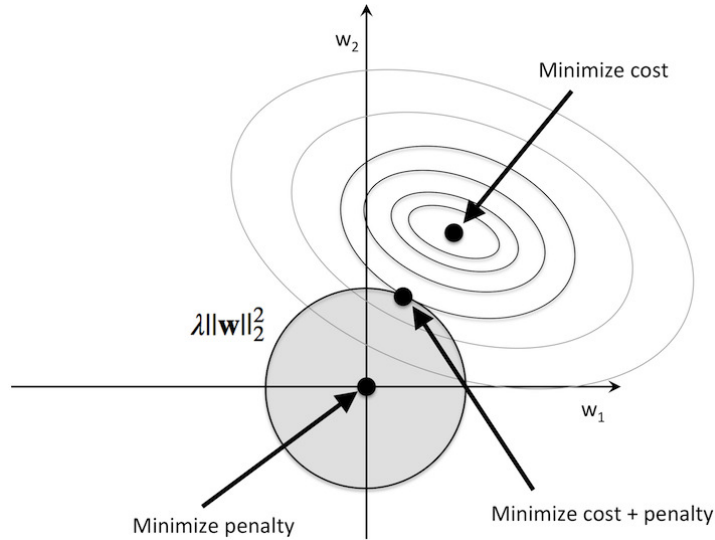


Figure 4.6: Graphical representation of ℓ_2 regularization

Here, if $\lambda = 0$ then we will get back our Linear regression model and if λ is too large, then it will add too much weight which will lead to a under-fit model. So, it is important to

choose a perfect value for λ . The solution of the minimization problem 4.1 is regularized W 's:

$$W_\lambda = \min (||AW - b||_2^2 + \lambda ||W||_2^2)$$

4.7.1 General view of ℓ_2 regularization

The minimization problem 4.7 can be written in more general way

$$W_\lambda = \arg \min (||AW - b||_2^2 + \lambda ||L(W - W_0)||_2^2)$$

where, W_0 is initial estimate of W which is set to zero when prior knowledge about W is unavailable, L is a linear operator. The problem is in standard form if $L = I$, the identity matrix.

Still our regularization parameter λ is unknown, now we will describe a method of finding the λ .

4.7.2 L-curve method of finding λ

A log-log plot of the norm of a regularized solution ($||w_\lambda||_2$) vs the norm of the associated residual norm ($||Aw - b||_2$) is the L-curve. It is an excellent graphical tool for showing the trade-off between the size of a regularized solution and its fit to the given data, as the regularization parameter λ varies. As a result, the L-curve method provides insights about the regularizing properties of the underlying regularization method and it helps to find an appropriate regularization parameter for the given data.

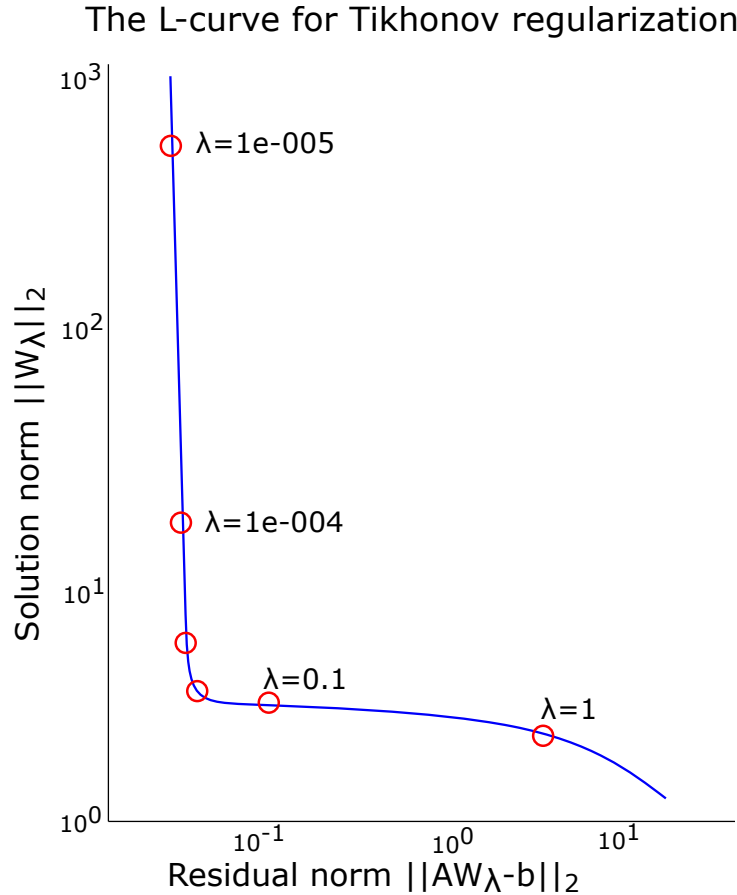


Figure 4.7: The L-curve for ℓ_2 regularization

To draw this above graph, we take different λ value and compute the pair $(\|Aw - b\|_2, \|w_\lambda\|_2)$ and plot it on the graph. And we will choose a λ such that it minimizes the equation 4.1.

The solution or appropriate λ value is the corner point of the L-curve.

4.8 ℓ_1 regularization

ℓ_1 regularization adds ℓ_1 norm of coefficient as penalty term or regularization term to the error function or objective function.

$$\min (\|AW - b\|_2^2 + \lambda \|W\|_1) \quad (4.2)$$

If lambda is zero, we will get back Linear regression model and if lambda is too large then that will make the coefficients tend to zero hence a under-fit model. Basically ℓ_1 regularization is very similar to ℓ_2 regularization. However, instead of a squared penalty term as in ℓ_2 , we penalize the model by the absolute weight coefficients. Our “budget” has “sharp edges,” which is the geometric explanation for why the ℓ_1 model induces sparsity, as shown in the figure below.

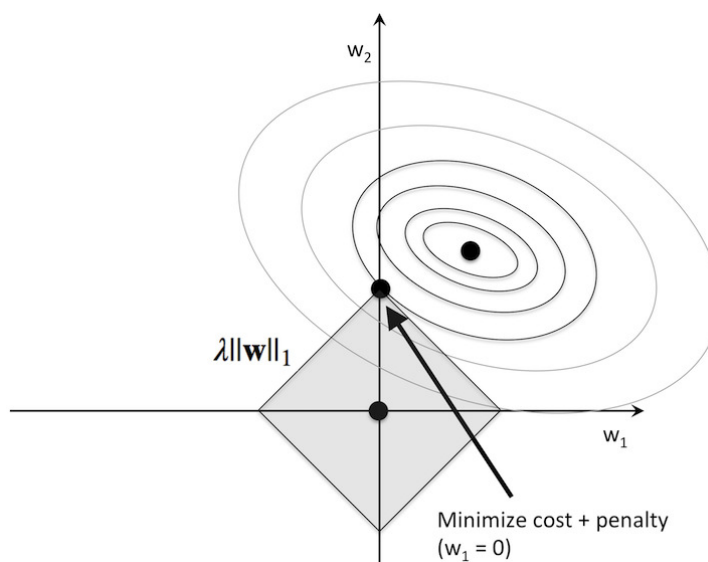


Figure 4.8: Graphical representation of ℓ_1 regularization

4.9 Differences between ℓ_1 and ℓ_2 regularization

In case of ℓ_1 regularization ℓ_1 norm of the coefficient is added to the objective function as a penalty term, whereas in ℓ_2 regularization, ℓ_2 norm is used.

By reducing the coefficient of the less significant feature to zero, the lasso completely eliminates some feature. In the event that we have a large number of features, this works well for feature selection. But ℓ_2 regularization does not completely eliminate any features instead it reduces the the importance of less significant feature by reducing the coefficient value.

Conclusion

In this report I reviewed most frequently used machine learning algorithms such as linear regression, logistic regression. The applications of these algorithm have been discussed along with the comparison between algorithms(whenever possible). We have solved some real life examples of linear regression and logistic regression.

In the first chapter, we have briefly discussed about machine learning, it's scopes and limitations. And we have also talked about supervised and unsupervised learning along with their classifications. And then we have given a short discussion about simple and multiple linear regression but in the next chapter we have discussed these two topics elaborately. At the end of this chapter, in short we have discussed about non-linear regression along with a nice graph.

In the second chapter, we have built a linear regression model and solved the model, that is, figuring out the parameters of the model, using direct method and gradient descent method. And we have tried to figure out the accuracy and relative error of the model. At the end, we have solved some real life examples, from that, we can conclude that, linear regression model best fits the data-set when the dependent variable and independent variable(s) are linearly related.

In the next chapter, we have built a logistic regression model for classification problem, we have discussed here, why linear regression is not best for classification problem and then we have discussed two important terms of logistic regression: threshold value and decision boundary(linear and non-linear), at the end, we have discussed about some real life examples that can be best explained by logistic regression model.

In the final chapter, we have talked about regularization and how we can obtain a good-fit model from an over-fit or under-fit model using two different regularization techniques: ℓ_1 and ℓ_2 regularization. We have briefly discussed about what we mean by bias and variance in regularization and at the end we have show what are the differences between ℓ_1 and ℓ_2 regularization.

After doing this project, it is anticipated that it will provide me with the knowledge that I need to make an informed choice when determining the options for machine learning algorithms that are available and then choosing the best machine learning algorithm for the particular problem-solving situation.

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